

EJERCICIOS EN OCTAVE

Ejercicio 1 y 2

```
clc;

[x,y]=meshgrid(-4:1:4,-4:1:4);

% --- 1 ---
%---ej 1
u1 = (1/(2*pi))*(x./(x.^2+y.^2));
v1 = (1/(2*pi))*(y./(x.^2+y.^2));
figure(1);
quiver(x,y,u1,v1,'b');
hold on;
%----ej 2
% c = 0;
% u1 = (c/r) = 0/r = 0;
u2 = zeros(size(x));
% v1 = 0;
v2 = zeros(size(x))
quiver(x,y,u2,v2,'r')
title('Potencial y Corriente 1');
axis tight;
xlabel("x");
ylabel("y");

%--- 2 ---
%----ej 1
u2 = 1;
u2 = ones(length(x),length(y));
v2 = 0;
v2 = zeros(length(x),length(y));
figure(2);
quiver(x,y,u2,v2,'b');
hold on;
%----ej 2
u2 = ones(length(x),length(y));
v2 = zeros(length(x),length(y));
quiver(x,y,u2,v2,'r')
title('Potencial y Corriente 2');
axis tight;

% --- 3 ---
%---ej1
%definicion de la funcion
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A = 1;
n = 2;
r = x.^(2) + y.^(2);
theta = atan2(y,x);
u_r = A*n*(r.^(n-1))*cos(n*theta);
u_theta = -A*n*(r.^(n-1))*sin(n*theta);
u = u_r*cos(theta) - u_theta*sin(theta);
v = u_r*sin(theta) + u_theta*cos(theta);
figure 3;
quiver(x,y,u,v,'b');
hold on
title('Potencial y Corriente 3');
axis tight;

%---ej2
u_r = A*n*(r.^(n-1))*cos(n*theta);
u_theta = -A*n*(r.^(n-1))*sin(n*theta);

u = u_r*cos(theta) - u_theta*sin(theta);
v = u_r*sin(theta) + u_theta*cos(theta);
quiver(x,y,u,v,'r');

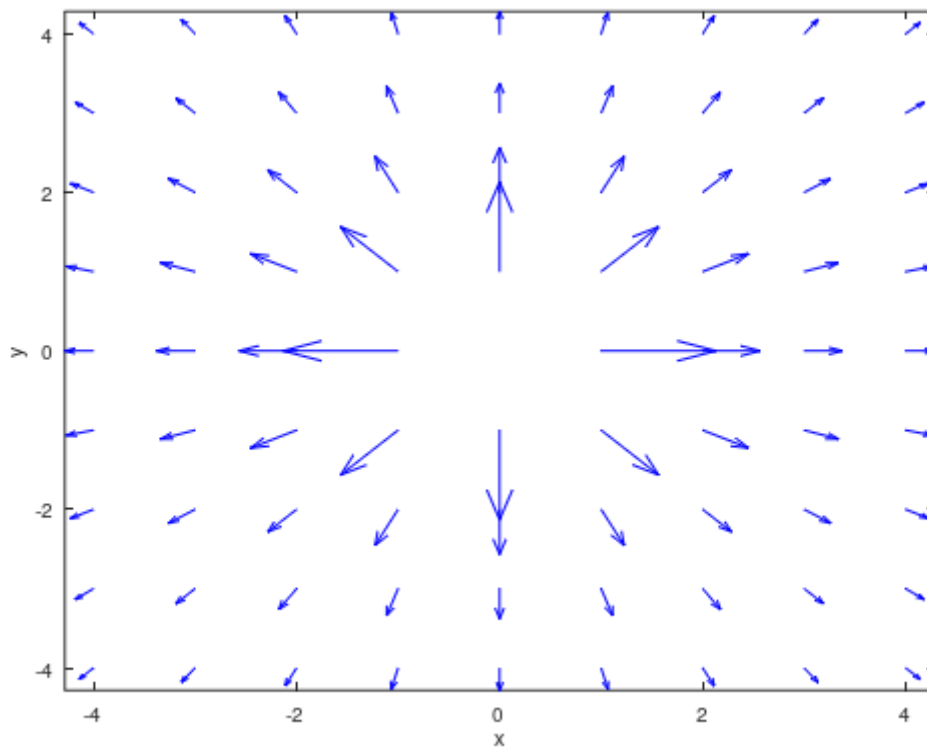
## % --- 4 ---
%---ej1
u_r = -cos(theta)./(r^(2));
u_theta = -sin(theta)./(r^(2));
u = u_r*cos(theta) - u_theta*sin(theta);
v = u_r*sin(theta) + u_theta*cos(theta);
figure 4;
quiver(x,y,u,v,'b');
hold on;
title('Potencial y Corriente 4');
axis tight;

%--ej2
u_r = -cos(theta)./(r^(2));
u_theta = -sin(theta)./(r^(2));
u = u_r*cos(theta) - u_theta*sin(theta);
v = u_r*sin(theta) + u_theta*cos(theta);
quiver(x,y,u,v,'r');

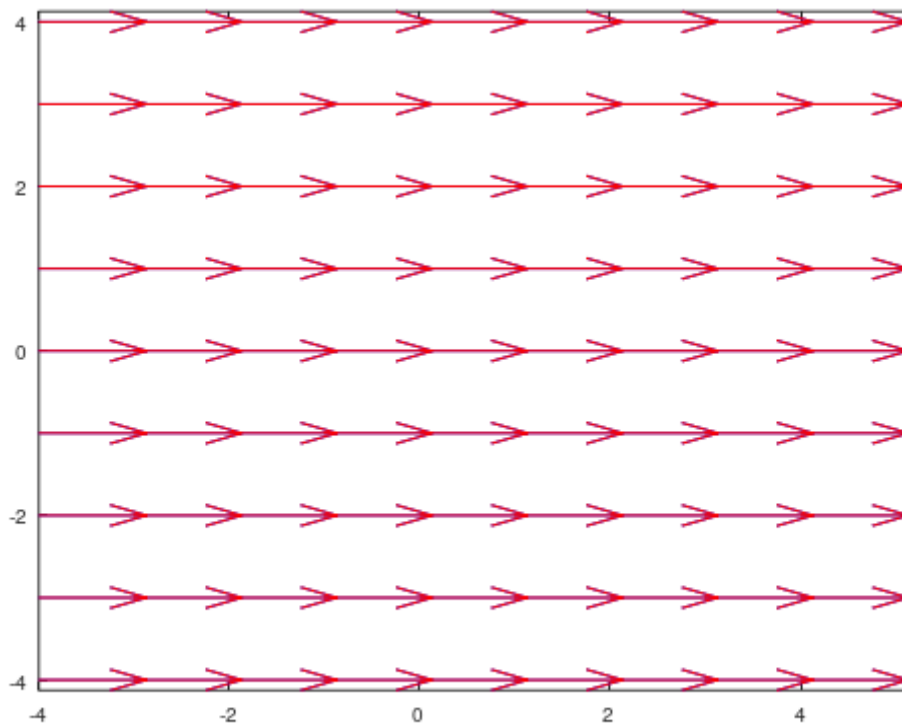
```

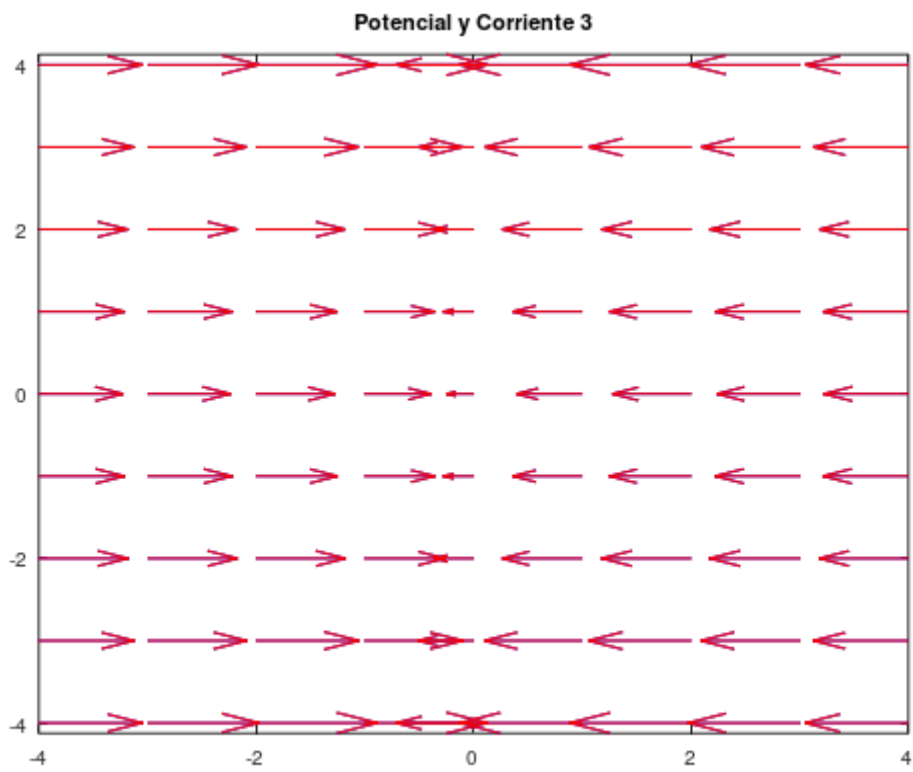
Graficas de potencial y corriente

Potencial y Corriente 1

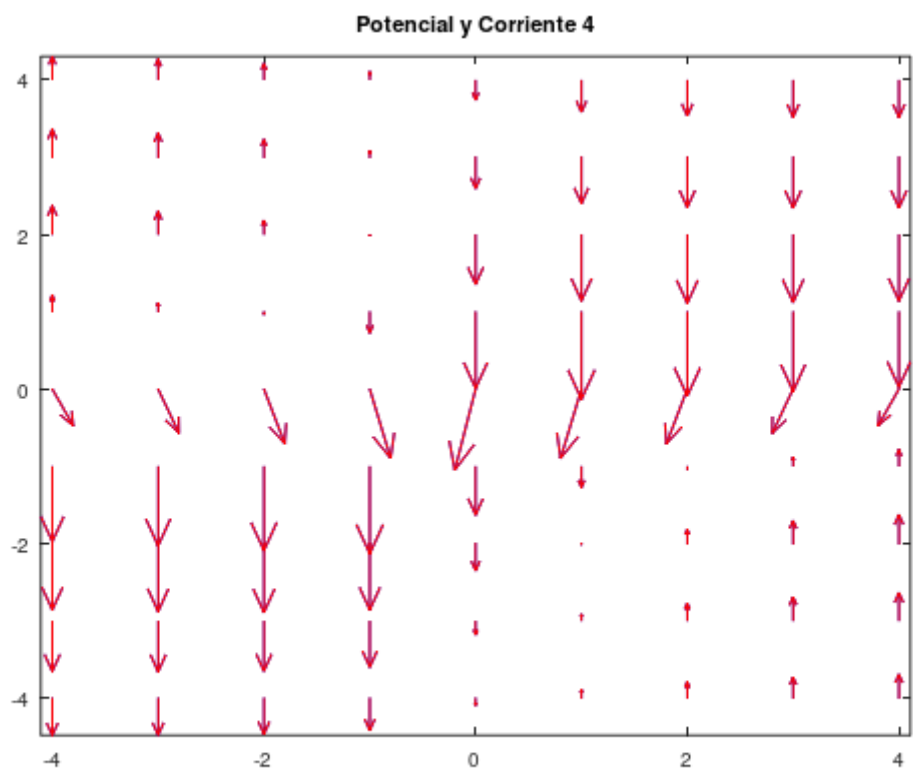


Potencial y Corriente 2





$(-3.941, 3.7168)$



$(-3.3471, 4.2839)$

Ejercicio 3 c)

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pkg load symbolic;

% Verificación de divergencia:
% Si  $\text{div}(\mathbf{v}) = 0$ , entonces el flujo es incompresible

syms x y r theta A n c

% Potenciales y sus divergencias
phi_a = (1/(2*pi))*log(r); % Calculado utilizando coordenadas polares
div_phi_a = diff(diff(phi_a, r), r) + (1/(r^2))*diff(diff(phi_a, theta), theta) + (1/r)*(diff(phi_a, r));

phi_b = x;
div_phi_b = diff(diff(phi_b, x), x) + diff(diff(phi_b, y), y);

phi_c = A*(r^n)*cos(n*theta);
div_phi_c = diff(diff(phi_c, r), r) + (1/(r^2))*diff(diff(phi_c, theta), theta) + (1/r)*(diff(phi_c, r));

phi_d = cos(theta)/r;
div_phi_d = diff(diff(phi_d, r), r) + (1/(r^2))*diff(diff(phi_d, theta), theta) + (1/r)*(diff(phi_d, r));

% Funciones de corriente y sus divergencias
psi_a = c*theta;
div_psi_a = diff(diff(psi_a, r), r) + (1/(r^2))*diff(diff(psi_a, theta), theta) + (1/r)*(diff(psi_a, r));

psi_b = y;
div_psi_b = diff(diff(psi_b, x), x) + diff(diff(psi_b, y), y);

psi_c = A*(r^n)*sin(n*theta);
div_psi_c = diff(diff(psi_c, r), r) + (1/(r^2))*diff(diff(psi_c, theta), theta) + (1/r)*(diff(psi_c, r));

psi_d = -sin(theta)/r;
div_psi_d = diff(diff(psi_d, r), r) + (1/(r^2))*diff(diff(psi_d, theta), theta) + (1/r)*(diff(psi_d, r));

% Mostrar los resultados
disp('div_phi_a =');
disp(div_phi_a);

disp('div_phi_b =');
disp(div_phi_b);

disp('div_phi_c =');
disp(div_phi_c);

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disp('div_phi_d =');
disp(div_phi_d);

disp('div_psi_a =');
disp(div_psi_a);

disp('div_psi_b =');
disp(div_psi_b);

disp('div_psi_c =');
disp(div_psi_c);

disp('div_psi_d =');
disp(div_psi_d);

```

Ejercicio 5

```

pkg load symbolic;
clear;
close all;

%-----
%----Inciso a---
%-----

% Declaración de variables
syms a r theta c z;
syms e_r_r e_theta_theta e_r_theta e_z_r e_z_theta e_z_z;

% Campo de desplazamiento
u_r = a*r*log(theta);
u_theta = a*r^2 + c*sin(theta);
u_z = 0;

% Componentes del tensor
e_r_r = diff(u_r,r);
e_theta_theta = (u_r / r) + (1 / r)*diff(u_theta,theta);
e_r_theta = (1/2)*((1/r)*diff(u_r,theta) + diff(u_theta,r) - (u_theta / r));
e_z_r = (1/2)*(diff(u_r,z) + diff(u_z,r));
e_z_theta = (1/2)*((1/r)*diff(u_z,theta) + diff(u_theta,z));
e_z_z = diff(u_z,z);

% Verificación de compatibilidad
szz = (2/r)*diff(diff(e_r_theta,theta),r) -...
      diff(diff(e_theta_theta,r),r) -...
      (1/(r^2))*diff(diff(e_r_r,theta),theta) +...
      (1/r)*diff(e_r_r,r) +...

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        (2/(r^2))*diff(e_r_theta,theta) -...
        (2/r)*diff(e_theta_theta,r);

f_szz = function_handle(szz);

a = 1;
c = 1;
r = 1;

theta = linspace(1,pi/2,2*pi-1); % 0 < theta < 2*pi
valor_szz = f_szz(a,c,r,theta) % Debe ser 0 para que el campo sea compatible

%---Inciso b---
%-----
% Gráfico del campo de desplazamiento

rango = linspace(-3,3,50);
[x,y] = meshgrid(rango,rango);
r = x.^2 + y.^2;
theta = atan2(y,x);
a = 1;
c = 1;

u = a*r*log(theta)*cos(theta) + a*(r.^2)*sin(theta) + c*(sin(theta).^2);
v = -a*r*log(theta)*sin(theta) + a*(r.^2)*cos(theta) + c*sin(theta)*cos(theta);

% Gráfico
figure(1);
quiver(x,y,u,v);
title('Campo de Desplazamiento');
xlabel("x");
ylabel("y");
axis tight;

```

Campo de desplazamiento ejercicio 5

Campo de Desplazamiento

